

Joint Transmitter-Receiver Design for High-Baud-Rate IM/DD Transmission with Low-Resolution DAC over 20-km Hollow-Core Fiber

Yibin Li⁽¹⁾, Jingchuan Wang⁽¹⁾, Zixian Wei⁽¹⁾, Qi Wu⁽¹⁾, Li Wang⁽¹⁾, Peng Li⁽²⁾, Lei Zhang⁽²⁾, Jie Luo⁽²⁾, Kangping Zhong⁽¹⁾, Alan Pak Tao Lau⁽¹⁾, Chao Lu⁽¹⁾, and Changyuan Yu⁽¹⁾

⁽¹⁾ Photonics Research Institute, Department of Electrical and Electronic Engineering, The Hong Kong Polytechnic University, Hong Kong SAR, China, jingchuan98.wang@connect.polyu.hk, changyuan.yu@polyu.edu.hk.

⁽²⁾ State Key Laboratory of Optical Fiber and Cable Manufacture Technology, Yangtze Optical Fiber and Cable Joint Stock Limited Company, Wuhan 430074, China.

Abstract We propose a joint transmitter-side VF-DRE and receiver-side NWF and MLSE scheme for high-baud-rate IM/DD transmission with low-resolution DAC. 140-GBaud OOK with 2-bit DAC and 100-GBaud PAM4 with 3-bit DAC are experimentally demonstrated over 20-km hollow-core fiber. ©2026 The Author(s)

Introduction

Recently, interest in high-baud-rate intensity-modulation/direct-detection (IM/DD) systems has revived with the emergence of hollow-core fiber (HCF) [1, 2]. This renewed attention is largely driven by the potential of HCF to enable ultra-wideband transmission while maintaining low fiber nonlinearity and ultra-low chromatic dispersion (CD). Moreover, for high-capacity transmission, a single high-speed link generally provides better cost efficiency and higher information density than multiple slower parallel links [3]. However, pushing IM/DD links to higher baud rates places stringent requirements on the transmitter digital-to-analog converter (DAC). On the one hand, high-resolution DAC generally comes at the expense of high cost and power consumption, while insufficient DAC resolution introduces severe quantization noise. On the other hand, the finite sampling rate of practical DAC hardware has become another fundamental bottleneck for high-baud-rate IM/DD links [4, 5]. To enable the implementation of low-resolution DAC, digital-resolution-enhancer (DRE) has been proposed to reshape quantization noise outside the signal band, thereby improving the in-band signal-to-quantization-noise ratio (SQNR) [6, 7]. More recently, Viterbi-free DRE (VF-DRE) has been introduced as a low-complexity alternative that preserves the key noise-shaping benefit of DRE while avoiding high-complexity Viterbi-based optimization [8].

However, for a high baud rate system, the benefits of VF-DRE vanish under a finite DAC sampling rate constraint. As the baud rate increases, the OSR decreases under a finite DAC sampling rate, and the signal band occupies a larger fraction of the DAC Nyquist band [9]. Consequently, more shaped quantization noise

falls back into the signal band, reducing the achievable SQNR gain. In addition, VF-DRE reshapes the quantization noise into a colored spectrum, which becomes more evident after receiver-side post equalization due to high-frequency noise enhancement [10, 11]. Therefore, the primary challenge lies in addressing the high-baud-rate performance degradation of VF-DRE from a colored-noise perspective, and recovering the associated performance loss through receiver-side noise whitening and sequence detection.

To overcome this challenge, this paper proposes a joint transmitter-receiver design for high-baud-rate IM/DD transmission over 20-km HCF with low-resolution DAC, where VF-DRE is employed at the transmitter and noise whitening filter (NWF) and maximum-likelihood sequence estimation (MLSE) are introduced at the receiver. Experimental results show that the proposed scheme significantly restores the performance gain of VF-DRE at high baud rates, enabling 140-GBaud OOK transmission with 2-bit DAC and 100-GBaud PAM4 transmission with 3-bit DAC under the 20% soft-decision forward-error-correction (SD-FEC) threshold.

Principle

For high-baud-rate IM/DD links, signal baud rates may approach or even exceed the available converter sampling rate with sub-sampling operation. Under a finite DAC sampling rate f_s , the oversampling ratio (OSR) with respect to the occupied signal bandwidth B_{sig} is

$$OSR = \frac{f_s}{2B_{sig}} \approx \frac{f_s}{(1+\beta)R_s}. \quad (1)$$

where β is the roll-off factor and R_s is baud rate. Denoting the quantization-noise power spectral density (PSD) after VF-DRE by

$$S_{VF}(f) = |H_{NS}(f)|^2 S_{RO}(f), \quad |f| \leq \frac{f_s}{2}, \quad (2)$$

where S_{RO} is the PSD of the round-off (RO) quantization noise and H_{NS} is the equivalent noise-shaping response, the in-band quantization-noise power after VF-DRE is

$$P_{VF} = \int_{-B_{sig}}^{B_{sig}} S_{VF}(f) df. \quad (3)$$

Introducing the normalized frequency $v = f/f_s$, this can be rewritten as

$$P_{VF} = f_s \int_{-B_{sig}/f_s}^{B_{sig}/f_s} S_{VF}(f_s v) dv, \quad (4)$$

Using the definition of OSR , we have

$$P_{VF} = f_s \int_{-\frac{1}{2OSR}}^{\frac{1}{2OSR}} S_{VF}(f_s v) dv. \quad (5)$$

Under a fixed DAC sampling rate, increasing the baud rate enlarges the occupied signal bandwidth and reduces the OSR . Since VF-DRE reshapes the quantization noise by shifting it toward high frequencies, a lower OSR causes more shaped noise to fall back into the signal band. Consequently, the in-band quantization noise power after VF-DRE increases. At extreme high baud rates, where the signal occupies most of the DAC Nyquist band, the SQNR gain of VF-DRE becomes even negligible.

This baud-rate-dependent degradation can be addressed from a colored-noise perspective. VF-DRE reshapes the quantization noise into a colored spectrum, which becomes more pronounced after post equalization. Therefore, instead of relying on transmitter-side VF-DRE alone, the degraded performance can be further recovered by receiver-side colored noise mitigation. Based on this viewpoint, a NWF is introduced to whiten the colored noise and MLSE is employed for decoding.

Experimental Setup

A proof-of-concept experiment was carried out using high-baud-rate OOK and PAM4 signals.

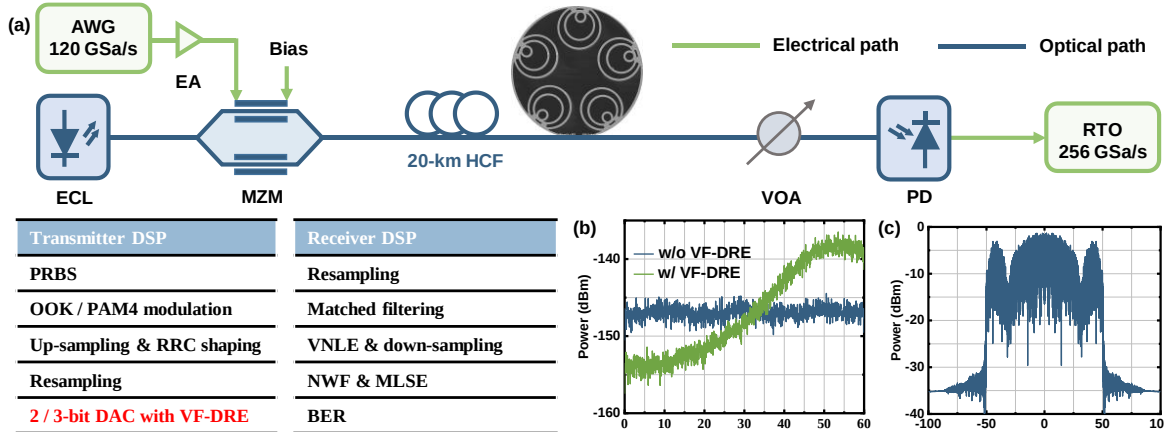


Fig. 1: (a) Experimental setup and DSP for IM/DD system. (b) The PSDs of quantization noise at a fixed AWG sampling rate of 120 GSa/s. (c) The electrical spectra of the received 100-Gbaud PAM4 signal after 20-km HCF transmission.

The experimental setup and the corresponding DSP flow are illustrated in Fig. 1(a). At the transmitter, a pseudo-random binary sequence (PRBS) was mapped into OOK or PAM4 symbols, followed by upsampling to 2 samples per symbol (sps) and pulse shaping with a root-raised-cosine (RRC) filter with a roll-off factor of 0.01. The generated waveform was then resampled to 120 GSa/s and quantized by VF-DRE. 2-bit and 3-bit DACs are used for OOK and PAM4 signals, respectively. The quantized signal is loaded into a 120-GSa/s arbitrary waveform generator (AWG, Keysight M8194A). The electrical output was amplified by a broadband linear electrical amplifier (EA, SHF S807C) with 23-dB gain to drive a Mach-Zehnder modulator (MZM, Fujitsu FTM7938EZ). An external-cavity laser (ECL) operating at 1550 nm was used as the optical carrier. The modulated optical signal was transmitted over 20-km HCF. At the receiver, the received optical power (ROP) was controlled by a variable optical attenuator (VOA). The signal was then directly detected by a 70-GHz photodiode and captured by a 256-GSa/s real-time oscilloscope (RTO, Keysight UXR0804A). In the offline receiver DSP, the sampled waveform was first resampled to 2 sps and passed through a matched filter. To compensate for the severe transceiver bandwidth limitation and device nonlinearity, a sparse Volterra nonlinear equalizer (VNLE) with 101 linear taps and 15 nonlinear taps was employed. The equalized sequence was then downsampled to 1 sps, followed by an NWF and MLSE for symbol detection and BER evaluation.

Results and Discussion

Fig. 1(b) shows the power spectral density (PSD) of the quantization noise after VF-DRE under a fixed AWG sampling rate of 120 GSa/s. VF-DRE suppresses the low-frequency quantization noise, and shifts more noise to higher frequencies. However, the PSDs with and without VF-DRE

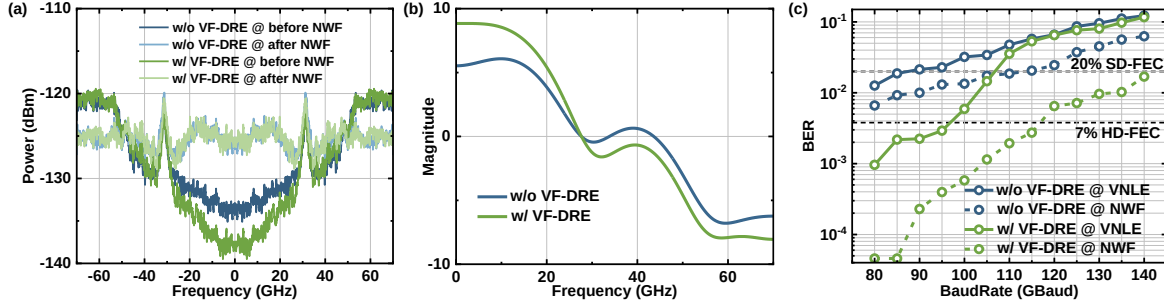


Fig. 2: Experimental results of OOK transmission with 2-bit DAC. (a) The colored noise spectrum before NWF and after NWF. (b) The frequency responses of NWF. (c) BER versus baud rate.

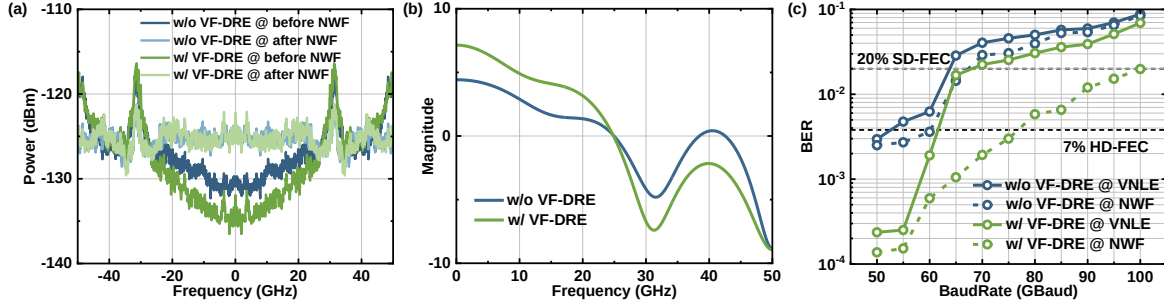


Fig. 3: Experimental results of PAM4 transmission with 3-bit DAC. (a) The colored noise spectrum before NWF and after NWF. (b) The frequency responses of NWF. (c) BER versus baud rate.

intersect at around 30 GHz. Therefore, as the baud rate increases beyond 60 GBaud, a larger portion of the shaped quantization noise falls into the signal band, and the gain provided by VF-DRE becomes increasingly limited. In addition, the received 100-GBaud PAM4 signal spectrum in Fig. 1(c) shows a spectral null around 30 GHz, which is induced by the dispersion of the 20-km HCF link.

For OOK transmission with 2-bit DAC, Fig. 2(a) shows the post-equalization noise spectrum for 140-GBaud OOK transmission. Before NWF, the case with VF-DRE shows more severe colored-noise enhancement, indicating that the high-frequency quantization noise becomes more pronounced after equalization. After applying NWF, most of the colored noise is effectively whitened for both cases. Nevertheless, a residual peak remains around 30 GHz, which is attributed to the dispersion-induced spectral null of the 20-km HCF link. The frequency responses of NWF in Fig. 2(b) show that a stronger high-frequency suppression is required for the case with VF-DRE. The BER curves in Fig. 2(c) further confirm that with VNLE only, VF-DRE provides a clear gain at relatively low baud rates, but this gain gradually decreases as the baud rate increases, and becomes negligible beyond 110 GBaud. When NWF and MLSE are introduced, the BER is significantly reduced, particularly for the VF-DRE case over the whole baud-rate range from 80 to 140 GBaud, enabling 140-GBaud OOK transmission with 2-bit DAC while satisfying the 20% SD-FEC.

For PAM4 transmission with 3-bit DAC, similar trends are observed in Fig. 3. In Fig. 3(a), the post-equalization noise becomes more strongly colored when VF-DRE is applied, and Fig. 3(b) also shows that the NWF designed for the VF-DRE case requires stronger suppression in the high-frequency band. The BER results in Fig. 3(c) show that the gain of VF-DRE also decreases with baud rate for PAM4. Below 60 GBaud, VF-DRE provides noticeable BER improvement after VNLE. However, as the baud rate increases beyond 60 GBaud, the gain provided by VF-DRE decreases rapidly. The joint use of transmitter-side VF-DRE and receiver-side NWF and MLSE can significantly restore the performance advantage of VF-DRE, supporting 100-GBaud PAM4 transmission with 3-bit DAC. Therefore, receiver-side colored noise mitigation is essential for recovering the benefit of transmitter-side VF-DRE in high-baud-rate transmission with low-resolution DAC.

Conclusions

In this paper, the limitations of VF-DRE at high baud rates under a finite DAC sampling-rate constraint were analyzed in terms of reduced OSR and enhanced post-equalization colored noise. By combining transmitter-side VF-DRE with receiver-side noise whitening and sequence detection, we experimentally demonstrated 140-GBaud OOK with 2-bit DAC and 100-GBaud PAM4 with 3-bit DAC over 20-km HCF.

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